1.10 Quadratics: The Discriminant

Chapter 2 · Original slide 35 · Starter

How many distinct real solutions do each of the following have?

$x^2 - 12x + 36 = 0$

$x^2 + x + 3 = 0$

$x^2 - 2x - 1 = 0$

Chapter 2 · Original slide 36 · 5: The Discriminant

$ax^2 + bx + c = 0$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Note: Roots of a function f are the values of x such that $f(x) = 0$.

Similarly the roots of an equation are solutions to an equation in the form $f(x) = 0$

Looking at this formula, when in general do you think we have:

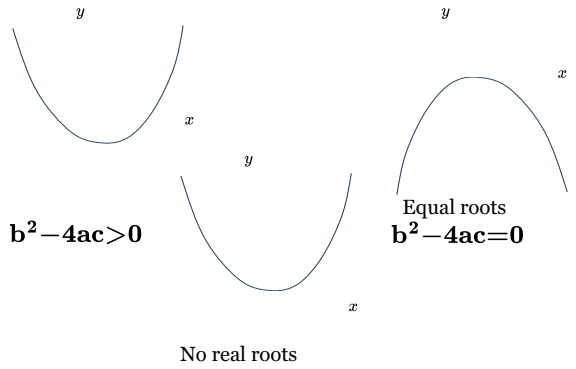
- No real roots?
- Equal roots?
- Two distinct roots?

$b2 - 4ac$is known as the discriminant.

Because adding 0 or subtracting 0 in the quadratic formula gives the same value.

1.10 Quadratics: The Discriminant

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We say the graph “touches” the axis.

Distinct real roots

$b^2-4ac<0$

Chapter 2 · Original slide 38 · Just for your interest…

Why do we say “Equal Roots” not “One root”?

It is due to the Fundamental Theorem of Algebra:

“Every polynomial of order n has exactly n roots.”

A polynomial is an expression with non-negative integer powers of x, i.e.

$a + bx + cx^2 + dx^3 + \ldots$

All linear, quadratic and cubic expressions are examples of polynomials.

$x^2 - 12x + 36 = 0$

Using the quadratic formula gives us the same value in both $+$ and $-$ cases: $x = 6$.

You might wonder why we say “*it has one repeated root*” or “*it has equal roots*”, i.e. indicating we have 2 roots (but with the same value). Why not say it has 1 root?

Despite the theorem being a simple statement, it was only until 1806 that it was first proven by Argand. Clearly by using the quadratic formula we can show a quadratic equation has 2 roots. We can use similar formulas to show that a cubic has 3 roots and a quartic 4 roots. But there is provably no such formulae for order 5 (quintics) and beyond. So we have to prove for example that 5 roots exist for a quintic, despite us having no way to find these exact roots!

One side result of the Fundamental Theorem of Algebra is that every polynomial can be written as a product of linear and/or quadratic expressions.

Leibniz claimed in 1702 that a polynomial of the form $x^4 + a^4$ cannot be written in this way. He then got completely burned by Euler in 1742 who managed to do so:

$$x^4 + a^4 = \left(x^2 + a\sqrt{2}x + a^2\right)\left(x^2 - a\sqrt{2}x + a^2\right)$$

The order of a polynomial is its highest power of x. So the order of a quadratic is 2, and a cubic 3.

1.10 Quadratics: The Discriminant

Chapter 2 · Original slide 39 · Quickfire Questions

Equation	Number of Distinct Real Roots
$x^2 - 4x + 1 = 0$	
$x^2 - 4x + 4 = 0$	
$2x^2 - 6x - 3 = 0$	
$x - 4 - 3x^2 = 0$	
$1 - x^2 = 0$	

Equation	Discriminant
$x^2 + 3x + 4 = 0$	

Chapter 2 · Original slide 40 · Problems involving the discriminant

8. The equation $x^2 + 2px + (3p + 4) = 0$, where p is a positive constant, has equal roots.

(a) Find the value of p. (4)

(b) For this value of p, solve the equation $x^2 + 2px + (3p + 4) = 0$. (2)

a)

Tip: Always start by writing out a, b and c explicitly.

b)

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2 Find the values of k for which $x^2 + 6x + k = 0$ has two real solutions.

3 Find the value of t for which $2x^2 - 3x + t = 0$ has exactly one solution.

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4 Given that the function $\mathrm{f}(x) = sx^2 + 8x + s$ has equal roots, find the value of the positive constant s.

5 Find the range of values of k for which $3x^2 - 4x + k = 0$ has no real solutions.

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7 $h(x) = 2x^2 + (k + 4)x + k$, where k is a real constant.

a Find the discriminant of $h(x)$ in terms of k.

b Hence or otherwise, prove that $h(x)$ has two distinct real roots for all values of k.

6 The function $g(x) = x^2 + 3px + (14p - 3)$, where p is an integer, has two equal roots.

a Find the value of p.

b For this value of p, solve the equation $x^2 + 3px + (14p - 3) = 0$.

Chapter 2 · Original slide 44 · Test Your Understanding

$x^2 + 5kx + (10k + 5) = 0$

where k is a constant.

Given that this equation has equal roots, determine the value of k.

Find the range of values of k for which $x^2 + 6x + k = 0$ has two distinct real solutions.

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Extension Questions:

1

[MAT 2009 1C] Given a real constant c, the equation

$x^4 = (x - c)^2$

Has four real solutions (including possible repeated roots) for:

A) $c \leq \frac{1}{4}$

B) $-\frac{1}{4} \leq c \leq \frac{1}{4}$

C) $c \leq -\frac{1}{4}$

D) all values of c

2

[MAT 2006 1B] The equation $\left(2 + x - x^2\right)^2 = 16$ has how many real root(s)?

3

[MAT 2011 1B] A rectangle has perimeter P and area A. The values P and A must satisfy:

A) $P^3 > A$

B) $A^2 > 2P + 1$

C) $P^2 \geq 16A$

D) $PA > A + P$